

Database Management System SE-208

LECTURE # 04

Database Architectures and the Web

Course Instructor: Engr. Ruqqiya Asad

OBJECTIVES

- The meaning of the client-server architecture and the advantages of this type of architecture for a DBMS.
- The difference between two-tier, three-tier and *n*-tier client-server architectures.
- About cloud computing and data as a service (DaaS) and database as a service (DBaaS).
- Software components of a DBMS.

OBJECTIVES

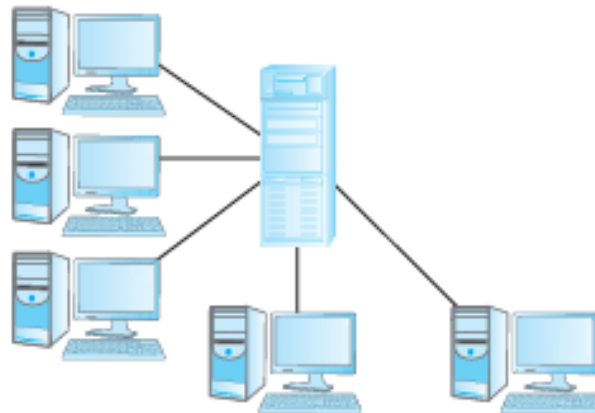
- The purpose of a Web service and the technological standards used to develop a Web service.
- The meaning of service-oriented architecture (SOA).
- The difference between distributed DBMSs, and distributed processing.
- The architecture of a data warehouse.
- About cloud computing and cloud databases.

Multi-user DBMS Architectures

- The common architectures that are used to implement multi-user database management systems:
 - Teleprocessing
 - File-Server
 - Client-Server

Teleprocessing

- One computer with a single central processing unit (CPU) and a number of terminals.
- All processing is performed within the boundaries of the same physical computer. User terminals are typically “dumb” ones, incapable of functioning on their own, and cabled to the central computer.

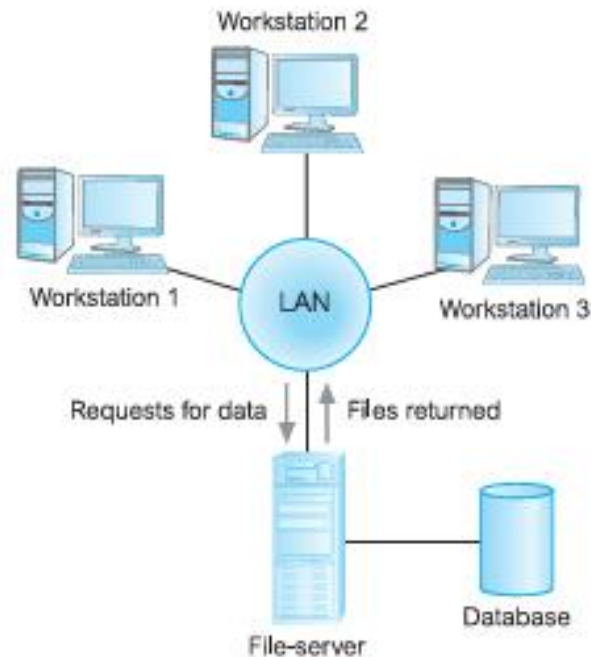


File-Server Architecture

- File-server is connected to several workstations across a network.
- Database resides on file-server.
- DBMS and applications run on each workstation.
- Disadvantages include:
 - Significant network traffic.
 - Copy of DBMS on each workstation.
 - Concurrency, recovery and integrity control more complex.

File-Server Architecture

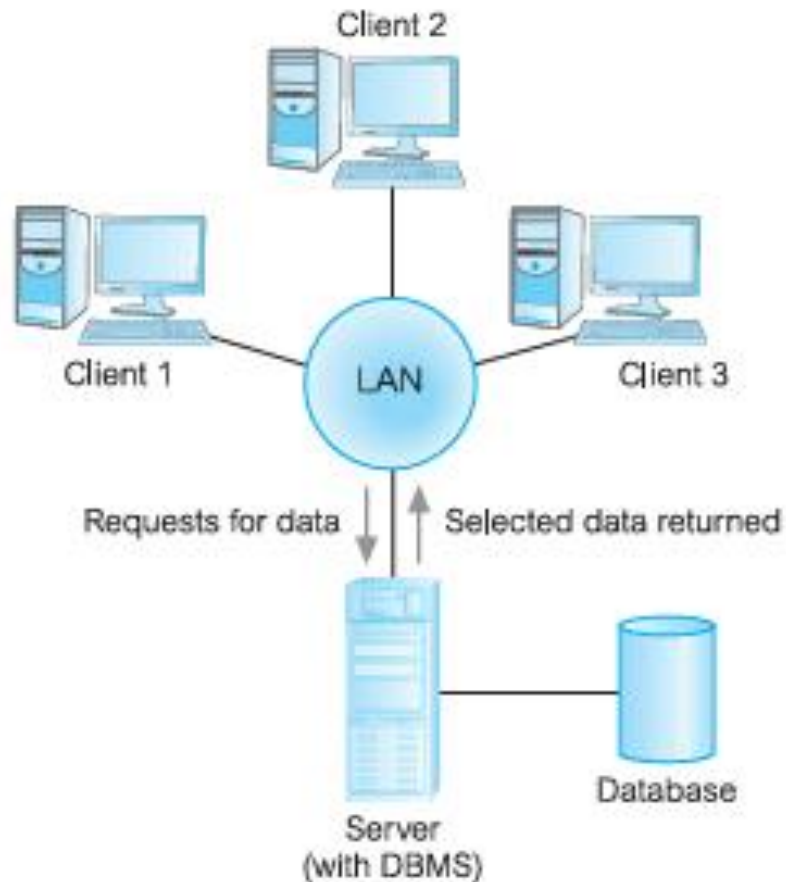
- In a file-server environment, the processing is distributed about the network, typically a local area network (LAN).



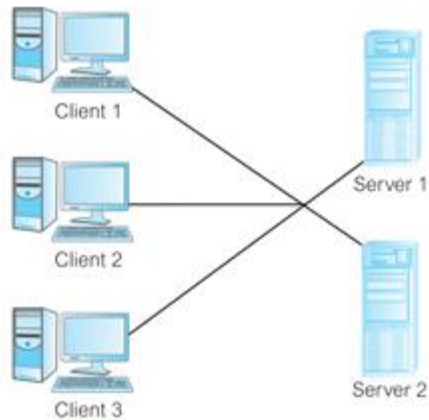
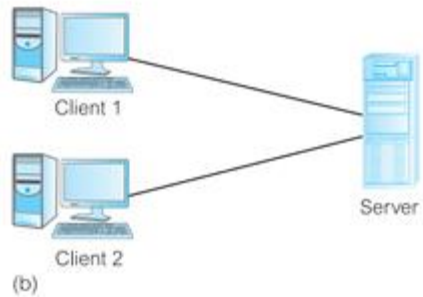
Traditional Two-Tier Client Server Architecture

- Client (tier 1) manages user interface and runs applications.
- Server (tier 2) holds database and DBMS.
- Advantages include:
 - wider access to existing databases;
 - increased performance;
 - possible reduction in hardware costs;
 - reduction in communication costs;
 - increased consistency.

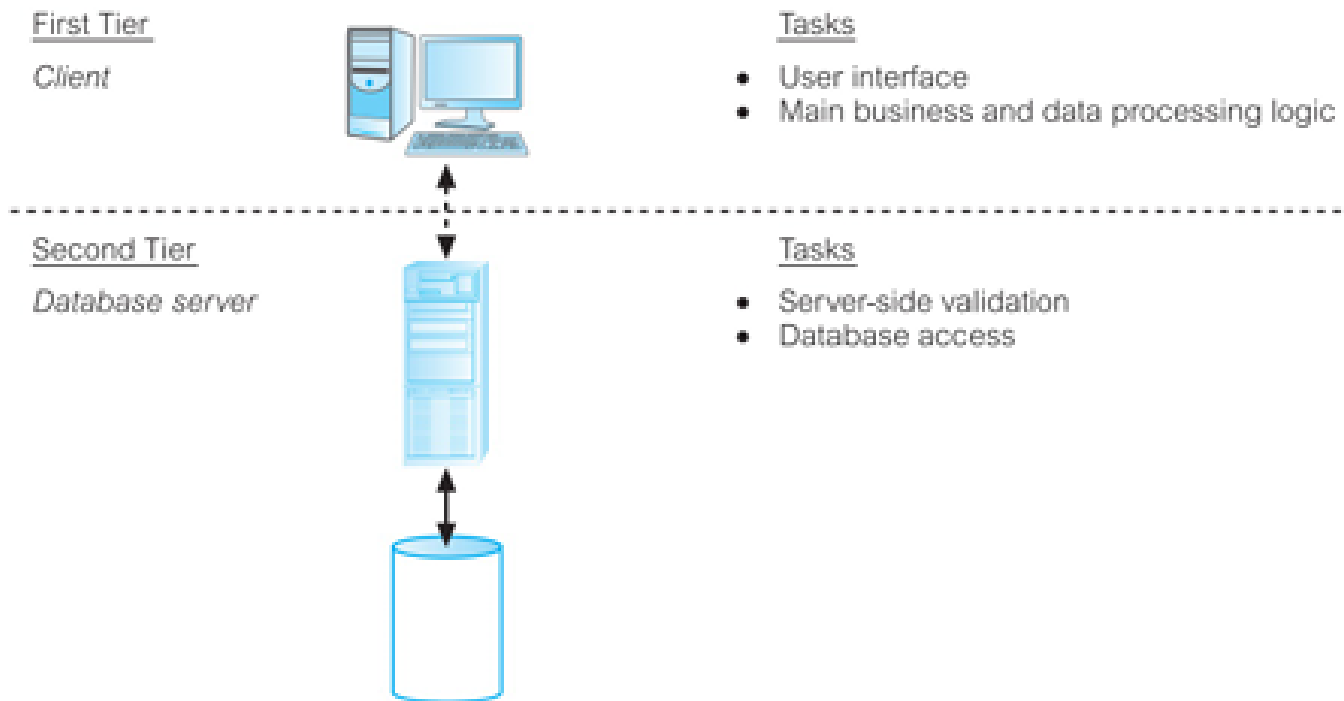
Traditional Two-Tier Client Server Architecture



Alternative client-server topologies



The traditional two-tier client–server architecture



Summary of client–server functions

CLIENT

Manages the user interface

Accepts and checks syntax of user input

Processes application logic

Generates database requests and transmits to server

Passes response back to user

SERVER

Accepts and processes database requests from clients

Checks authorization

Ensures integrity constraints not violated

Performs query/update processing and transmits response to client

Maintains system catalog

Provides concurrent database access

Provides recovery control

Three-Tier Client-Server

- The need for enterprise scalability challenged the traditional two-tier client-server model.
- Client side presented two problems preventing true scalability:
 - 'Fat' client, requiring considerable resources on client's computer to run effectively.
 - Significant client side administration overhead.
- By 1995, three layers proposed, each potentially running on a different platform.

Three-Tier Client–Server Architecture

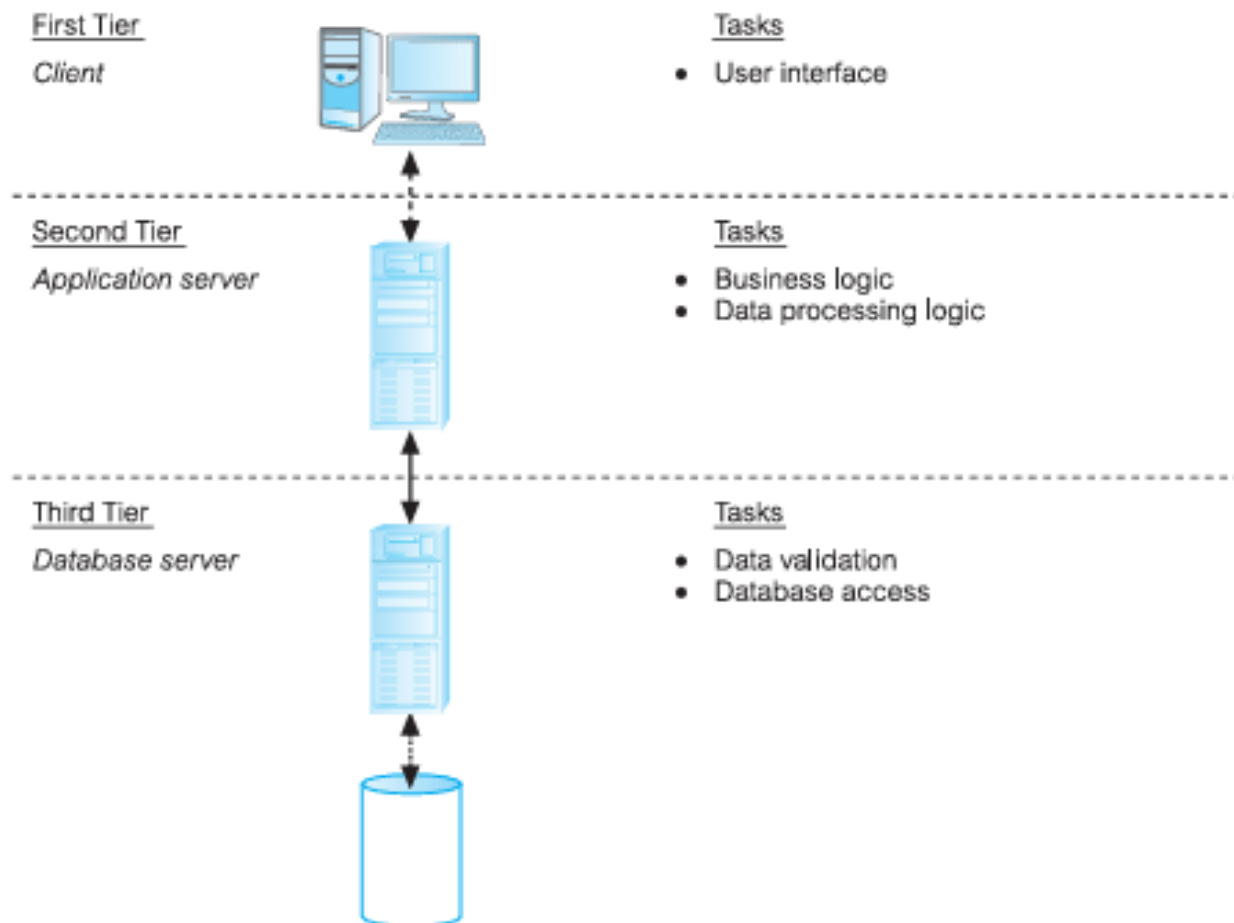
- The need for enterprise scalability challenged the traditional two-tier client–server model.
- Client side presented two problems preventing true scalability
 - 'Fat' client, requiring considerable resources on client's computer to run effectively.
 - Significant client side administration overhead.
- By 1995, three layers proposed, each potentially running on a different platform.

Three-Tier Client-Server

Advantages:

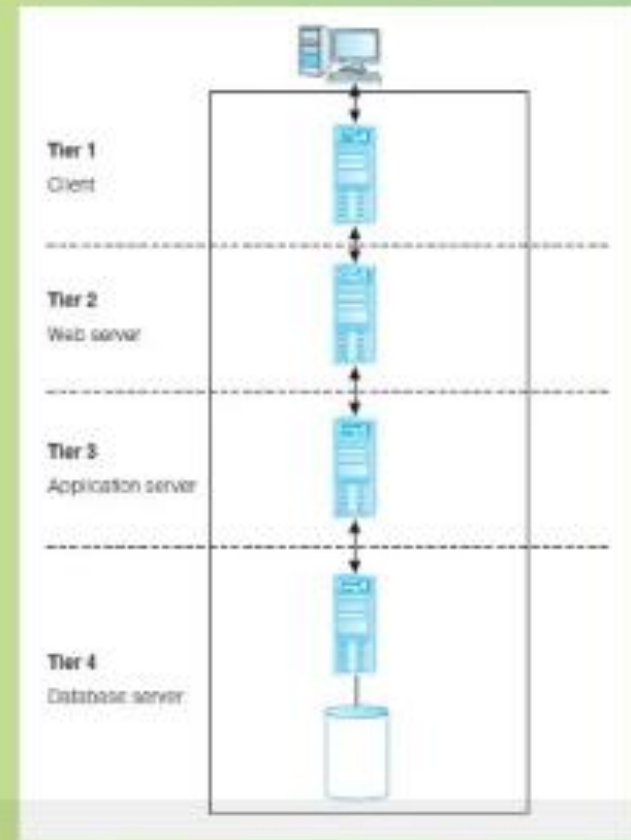
- 'Thin' client, requiring less expensive hardware.
- Application maintenance centralized.
- Easier to modify or replace one tier without affecting others.
- Separating business logic from database functions makes it easier to implement load balancing.
- Maps quite naturally to Web environment.

Three-Tier Client-Server Architecture



n-Tier Client-Server (e.g. 4-Tier)

- The three-tier architecture can be expanded to n tiers, with additional tiers providing more flexibility and scalability.
- Applications servers host API to expose business logic and business processes for use by other applications.



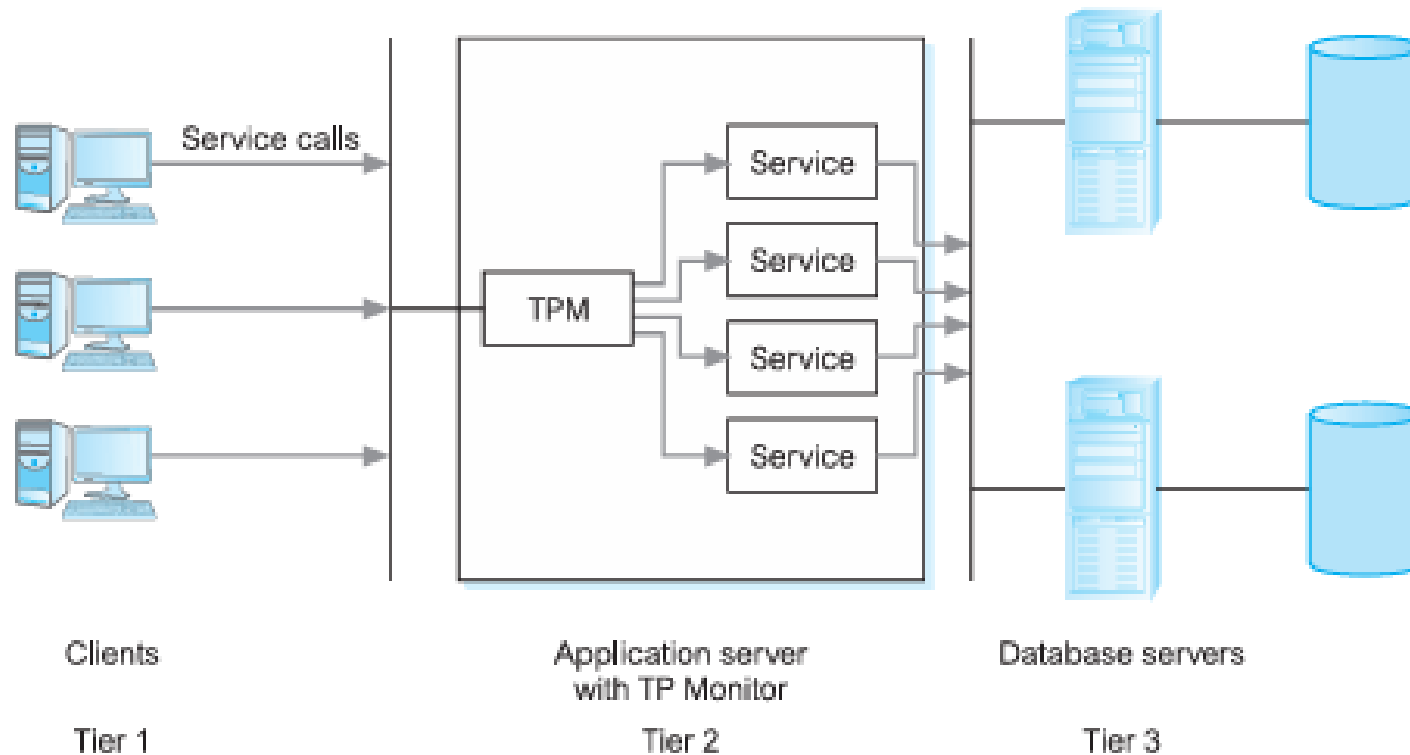
Middleware

- **Middleware is a generic term used to describe software that mediates with other software and allows for communication between disparate applications in a heterogeneous system.**
- **The need for middleware arises when distributed systems become too complex to manage efficiently without a common interface.**

Transaction Processing Monitors

- TP monitor is a program that controls data transfer between clients and servers in order to provide a consistent environment, particularly for online transaction processing (OLTP).

The Transaction Processing Monitor as the middle tier of a three-tier client-server architecture



Web Services and Service-Oriented Architectures

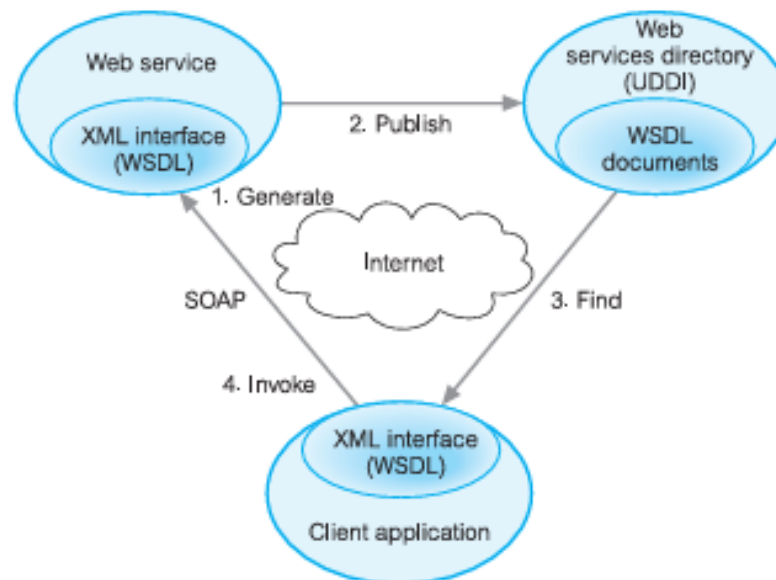
- Web service is a software system designed to support interoperable machine-to-web service machine interaction over a network.
- Web services share business logic, data, and processes through a programmatic interface across a network.
- Developers can add the Web service to a Web page (or an executable program) to offer specific functionality to users.

Web Services and Service-Oriented Architectures

- Web services approach uses accepted technologies and standards, such as:
 - XML (extensible Markup Language).
 - SOAP (Simple Object Access Protocol) is a communication protocol for exchanging structured information over the Internet and uses a message format based on XML. It is both platform- and language-independent.
 - WSDL (Web Services Description Language) protocol, again based on XML, is used to describe and locate a Web service.

Web Services and Service-Oriented Architectures

- UDDI (Universal Discovery, Description, and Integration) protocol is a platform independent, XML-based registry for businesses to list themselves on the Internet.



Service-Oriented Architectures (SOA)

- A business-centric software architecture for building applications that implement business processes as sets of services published at a granularity relevant to the service consumer. Services can be invoked, published, and discovered, and are abstracted away from the implementation using a single standards-based form of interface.

Distributed DBMSs

- A distributed database is a logically interrelated collection of shared data (and a description of this data), physically distributed over a computer network.
- A distributed DBMS is the software system that permits the management of the distributed database and makes the distribution transparent to users.

Distributed DBMSs

- A DDBMS consists of a single logical database split into a number of *fragments*.
- Each fragment is stored on one or more computers (*replicas*) under the control of a separate DBMS, with the computers connected by a network.
- Each site is capable of independently processing user requests that require access to local data (that is, each site has some degree of local autonomy) and is also capable of processing data stored on other computers in the network.

